Ben Bubnick

Data Engineer · Data Scientist · Analytics Specialist

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Summary

Hands-on engineer specializing in Python on UNIX-like systems (Linux/macOS), production ML/MLOps, and large-scale data pipelines. Heavy daily use of Terminal/Bash/Zsh, SSH, and POSIX tooling to build and operate services. Built forecasting and optimization solutions (e.g., surgery duration prediction, hospitalization risk) and led end-to-end data platforms in AWS and Azure. Strong at turning ambiguous operational problems into real-time services with CI/CD, observability, and documentation. Comfortable collaborating across engineering, data science, and ops—ready to lead AI application development and model integration in high-performance, resource- and cost-sensitive environments.

Experience

GrowData Analytics

AWS: S3, Glue, Athena, Lambda, CloudWatch, dbt, Snowflake, Ruby, RubyLLM, Tesseract OCR, GPT 4o mini, Cladue Opus/Sonnet, Rubocop, RSpec

Principal Data Engineer *Oct. 2025 - Present*

Built and shipped an OCR-to-analytics product for small farms; owned model integration, CI/CD, testing, and auditability on UNIX-like systems.

- Led end-to-end design and delivery of HHAX healthcare data integration pipelines, converting operational source data into curated analytics-ready models for reporting and downstream use.
- Conducted feasibility analysis of source-system structures and implementation constraints to shape integration scope, modeling strategy, and delivery approach.
- Designed and refined core domain models for organization, office, caregiver, patient, and crosswalk entities to support standardized healthcare data workflows.
- Structured target models using FHIR-like concepts where appropriate, improving consistency and extensibility for downstream integration and analytics use cases.
- Developed reusable ETL and transformation logic across development and sandbox environments using SQL and warehouse-native modeling practices.
- Implemented source-to-target standardization patterns for null handling, lifecycle fields, sentinel dates, and source-specific variation.
- Defined surrogate key, natural key, and crosswalk strategies to enable stable identity resolution across HHAX and related source systems.
- Designed and executed comprehensive QA and data-integrity test packs covering row counts, duplicates, nulls, referential integrity, reconciliation, and exception handling.
- Resolved data-quality and mapping issues through targeted test-case analysis and review of failing source and target records.
- Documented mapping logic, model assumptions, QA expectations, and implementation standards to improve delivery reliability and reuse.
- Collaborated with client and internal stakeholders to resolve source ambiguity, refine requirements, and translate business needs into production-ready data assets.
- Supported future scalability by establishing reusable patterns for onboarding additional sources and extending integration workflows.

Principal ML Engineer *Sep. 2024 - Oct. 2025*

Built and shipped an OCR-to-analytics product for small farms; owned model integration, CI/CD, testing, and auditability on UNIX-like systems.

- Productized OCR → structured data pipeline for farm invoices (handwritten/printed) using Tesseract + rule-based post-processing; shipped as a small Python/Ruby service with containerized deployments and versioned configs.
- Built validation & reconciliation layers (schema checks, cross-field rules, idempotent re-runs) and produced audit logs to support bookkeeping/compliance workflows.
- Implemented CI/CD (unit tests, linting, fixtures, golden-file tests) and release automation; documented runbooks and failure modes (timeouts, low-confidence OCR, retries).
- Delivered a lightweight analytics layer (usage metrics, error taxonomy, latency/throughput dashboards) to guide feature tuning and cost control.
- Owned infra as code for a minimal AWS footprint (S3 for artifacts, Lambda/Cron for jobs, CloudWatch alerts); operated via UNIX-like CLI workflows (Bash/Zsh, Makefile targets).

Health Data Movers

AWS: S3, Snowflake, SQLMesh, Python, SQL, Docker

Data Consultant *Mar. 2025 - Present*

Developed a reusable ETL pipeline to streamline data onboarding for a healthcare revenue cycle platform.

- Built a reusable ingestion & transformation framework in SQLMesh/Snowflake (templated models, data contracts, CI checks, promotion gates), enabling consistent, repeatable onboardings.
- Added observability & auditability (row-count/parity checks, schema drift alerts, reconciliation logs) and wrote runbooks for deterministic re-runs and incident handling on UNIX-like systems.
- Shipped a lightweight forecasting/alerting pattern (Python service + Docker + Actions) for operations signals; packaged with tests and deployment automation.
- Created testable, modular SQLMesh models, reducing regression errors and accelerating iteration cycles.
- Standardized and documented transformation logic across clients in order to cut onboarding documentation.
- Established validation and audit checkpoints (e.g., row counts, null checks, referential integrity).
- Translated business requirements into technical specifications to improve delivery predictability.
- Built detailed integration playbooks to be used as the internal standard for onboarding new vendors and partners.

Arkos Health

AWS: S3, Athena, Redshift, Glue, dbt, PySpark, Docker + AWS CLI, Tableau, Jupyter

Senior Data Analyst *Jun. 2023 - Jan. 2025*

Built a scalable AWS data lake with automated CI/CD, metadata reporting, and integrated analytics tools.

- Stood up an AWS data lake with CI/CD (dbt + Git flows), cost-aware partitioning, and job scheduling; reduced deployment time 50% and improved query performance with targeted model refactors.
- Implemented monitoring/alerting & on-call runbooks (metrics, logs, failure taxonomies) to lower MTTR and improve data reliability.
- Mentored engineers on code reviews, testing, and pipeline design; authored internal standards that lifted team delivery quality.
- Integrated Jupyter into reporting, boosting report interactivity and increasing stakeholder satisfaction by 30%.
- Created comprehensive data lake documentation, reducing onboarding time for new team members by 40%.
- Implemented Jira and GitHub, enhancing task tracking and team coordination by 25% within three months.
- Enhanced cross-departmental collaboration, increasing data-driven decision-making by 15% in six months.
- Mentored junior team members, improving their performance metrics by 30% in six months.
- Led goal-setting sessions, improving goal alignment and tracking accuracy by 30% in the first quarter.
- Initiated stakeholder feedback sessions, improving deliverable relevance and accuracy by 25% in three months.

Merative *Azure: Data Factory, Blob Storage, Functions, SQL DB, Synapse Analytics, Python, DevOps*

SENIOR DATA SCIENTIST *Jul. 2022 - Mar. 2023*

Launched a new Azure platform to replace HDFS systems to modernize data infrastructure.

- Led HIPAA-compliant deidentification initiative across 200+ million patient records.
- Designed and implemented an automated testing framework for ETL pipelines for validation coverage.
- Validated and optimized Azure-based data workflows, ensuring continuity and fidelity during the transition from legacy HDFS systems to cloud architecture.
- Conducted end-to-end system testing for Azure Data Factory pipelines, Blob Storage configurations, and downstream Synapse integrations.
- Collaborated with engineering teams across disciplines to refine architectural designs.
- Documented test plans and platform migration procedures for improving onboarding for future QA engineers.
- Built regression test suites and data comparison tools, detecting schema drift and transformation errors early in the development lifecycle.
- Ensured compliance with data governance and security protocols to reduce audit risks and support HITRUST readiness.

IBM *Hadoop/HDFS/CDH: Spark, Hive & Impala, Kafka, Python, JRuby, SQL, Pig, Docker + Gradle*

SENIOR DATA SCIENTIST *Nov. 2021 - Jul. 2022*

Built AI tools while leading analytics delivery and large-scale data validation efforts.

- Created AI Surgical Scheduling Optimization tool currently evolving into new offering.
- Led delivery on tailored analytics and reporting project for client administrative data.
- Developed data migration validation analysis on 150+ billion patient records.
- Developed surgical smart case card using combined CNN/RNN model.

SENIOR DATA CURATION ARCHITECT *Dec. 2020 - Nov. 2021*

Led large-scale clinical data and NLP initiatives and delivering ML solutions.

- Developed clinical data integration of 100+ billion structured & semi-structured records.
- Developed topic modelling project on nearly 6 million semi-structured provider notes data.
- Curated 25 pages of workflow documentation and mentored new hires in AI tuning process.
- Created COVID-19 hospitalization prediction tool with Machine Learning algorithms.

DATA CURATION ARCHITECT *Nov. 2017 - Dec. 2020*

Led enterprise data integration initiatives and mentoring programs.

- Managed multi-org enterprise integration with over 70 interconnected data sources.
- Developed new data validation and review process that reduced total rework by 25%.
- Reduced mapping rework by 45% with through new knowledge-driven initiatives.
- Led mentoring program to increase 4,500 billable hours over two year period.

SOFTWARE DEVELOPER *Sep. 2016 - Nov. 2017*

Developed internal resources that reduced costs and improved onboarding efficiency.

- Reduced time spent on knowledge transfer by 80% by rewriting project documentation.
- Eliminated need for a $30,000 annual contract by developing data curation tool.
- Reduced cost-per-hire by $5,210 by re-developing education program.

DATA SCIENTIST *Nov. 2015 - Sep. 2016*

Led integration efforts and process automation initiatives.

- Managed team of 12 remote data science contractors for 36 integration projects.
- Created over 125 new knowledge base articles, making up 30% of the team's online content.
- Developed automated data integration tool to cut out 28% unnecessary development time.

Explorys
Hadoop/HDFS: Ruby, Pig, React, NodeJS, JavaScript, SQL, Oracle, Jenkins

DATA SCIENTIST *Jun. 2015 - Nov. 2015*

Built ETL pipelines on distributed data systems to support clinical data integration and population analytics.

- Developed cron-based widget to define blackout periods for client data pulls.
- Performed troubleshooting on production issues across multiple environments.
- Principal integration specialist for 10+ client database warehouses.

Multiple Institutions

ADJUNCT PROFESSOR *Aug. 2010 - May. 2019*

Taught physics & astronomy lecture series and lab courses.

- Consistently exceeded student evaluation benchmarks, surpassing institutional feedback goals by an average of 16% in 2017 and 17% in 2018, reflecting strong teaching effectiveness and student satisfaction.
- Improved student comprehension and course navigation by reorganizing online instructional materials, resulting in a 28% reduction in reported confusion and fewer clarification requests during office hours.
- Developed and published over 700 hours of recorded lectures and lab content, enhancing accessibility and supporting flexible, on-demand learning for both traditional and non-traditional students.
- Implemented Just-in-Time Teaching (JiTT) strategies for underperforming students, leading to an average grade improvement of 18% by the end of term.
- Designed interactive lab lectures based on the Interactive Lecture Demonstration (ILD) method, increasing student engagement and concept retention in physics instruction.
- Promoted collaborative learning through Peer Instruction and ILD techniques, effectively addressing real-time student misconceptions during physics lab sessions.

AmTrust Financial Services
VB.NET, ASP.NET, C#, CSS, Javascript, HTML5, SQL, PL/SQL

SOFTWARE ENGINEER *Mar. 2014 - Jun. 2015*

Developed MVC components and web forms for insurance policy software.

- Led development on core insurance policy software, serving as principal developer for a new product offering and contributing to two major architectural initiatives in 2015.
- Analyzed and optimized performance across 65+ ISO rating-based web forms using R, identifying bottlenecks and reducing page load times.
- Championed and implemented a unit testing framework for multi-tier MVC components, improving code quality and reducing deployment errors across the dev team.

PPG Industries
VB.NET, C#, SQL, C++, FORTRAN

COLOR SCIENTIST *Apr. 2012 - Jun. 2013*

Developed MVC components and web forms for insurance policy software.

- Improved color matching classification algorithm accuracy by 40% by developing and applying advanced numerical methods, enhancing product performance across automotive coatings.
- Developed cross-instrument correlation software to support a risk analysis initiative, enabling consistent color measurement across varying hardware platforms and improving QA reliability.
- Supported the creation of an industry-leading automated color matching facility, developing numerical tools and methods to improve formulation quality and pigment data analysis in high-variance OEM color environments.

Professional Recognition

HDM, SPECIAL RECOGNITION (*2025*), Recognized for exceptional performance and commitment in client delivery.

Arkos, ARKOS HEALTH BI STAR (*2023*), Recognition for Achieving Expeditioner Rank, Salesforce Trailhead.

Merative, MANAGER RECOGNITION FOR TRANSITION LEADERSHIP (*2022*), Recognized for systems migration efforts.

IBM, GOLD CHAMPION (*2018 - 2022*), Recognized for 1,477 classroom hours delivered and 43 certificates earned.

IBM, LAB SERVICES AWARD (*2018*), Recognized for coordinating internal teams to deliver solutions on time.

IBM, EMINENCE AND EXCELLENCE AWARD (*2017*), Recognized for developing a new data curation tool.

PPG, COLOR LAB AUTOMATION AWARD (*2012*), Recognized for expanding data collection and analysis reporting.

Talks

AI Hackathon 2020: Surgery Scheduling Optimization *Dec. 2020*
"An algorithm uses past utilization and current waitlist info to predict surgery duration. IBM

REACH Initiative: IBM Watson Care Insights *2019*
Analyze patient records and real-time data via HL7 feeds, integrate insights into physician's EHR workflow. IBM

Content & Data Analytics Learning Exchange *2018*
Introduction of implementation of machine learning algorithms using real-world health data analytics. IBM

2018 Innovation Research Interchange Fall Networks Conference *Sep. 2018*
Using NLP to extract insight from unstructured data and transform it from a data lake. IRI

Cross Team Training Series *2016*
A 7-part series on using Hadoop for data ingestion that shows advantages of data lakes. EXPLORYS

Publications

Reproducible Postgres load of Tuva seed datasets *Aug. 2025*
Reproducible Postgres load of Tuva seed datasets with robust validation. HSEA

Surgical Scheduling Optimization and Procedure Duration Prediction [a] *Dec. 2020*
Improvement on traditional fixed ratio methods for total procedure time estimation. IBM

Process and Requirements for CCD Data Integration *Oct. 2017*
Processes and requirements for overcoming the many difficulties with CCD data curation. IBM

AI Paint Formula Prediction from Spectroscopic Measurements [b] *May. 2015*
Naive bayesian color and effects matching algorithm of complex automotive paint formula. SHINYAPPS.IO

Characterization of Surface Coatings using RT Models for Color Matching *Feb. 2013*
Approximation of Chandrasekhar RT model for atmospheric scattering applied to color matching. USPTO

Massive Stellar Clusters in the Disk of the Milky Way Galaxy [c] *Dec. 2010*
Spectroscopic and Photometric analysis of two open clusters in the galactic plane. OHIOLINK ETD

VizieR Online Data Catalog: JHK photometry in Cluster [BDS2003] 107 *May. 2008*
Spectra were obtained 2005 June 1415 at the 3.0 m NASA Infrared Telescope Facility (IRTF) on Mauna Kea, Hawaii, using SpeX. VIZIER

Near-Infrared Spectroscopy of Candidates Members of the Galactic Cluster [BDS2003] 107 *Feb. 2008*
New near-infrared classification spectra for nine candidate members of the galactic cluster [BDS2003] 107 (Cluster 107). PASP

A Novel Method for Low-Toxic Photo-Etch Printing [d] *May. 2003*
Application of aquatint etching techniques to photo-etch procedures. MAPC

Education

2022	**PGCert.**, Data Science Professional[a]	*IBM*
2015	**PGCert.**, Data Science[b]	*Johns Hopkins University*
2010	**Master of Science**, Astrophysics[c]	*University of Cincinnati*
2004	**Bachelor of Fine Arts**, Printmaking[d]	*Ohio University*

Volunteer

HSEA

ETL Brigade *Feb. 2025 - May. 2025*

Built a reusable PostgreSQL data model to help community health organizations prototype analytics tools for claims, visits, and eligibility.

CSforAll

Instructor *Oct. 2018 - May. 2019*

Our goal was to provide computer science instruction in all of the District's high schools, with a special emphasis on students who do not have access to a computer or a stable Internet connection at home.

HIT in the CLE

Coach *Mar. 2017 - Mar. 2019*

We were using a data science competition to educate students understand the potential of a career in health information technology and to assist IT companies in Northeast Ohio in developing a talent pipeline.

Code for America

ETL Guild *Jan. 2015 - Jun. 2015*

The initiative was brought about to apply data science tools to improve and expand public record access.

T.C.P. World Academy

Tutor *Sep. 2008 - May. 2009*

Our goal was to teach students how to understand the value of their learning experience and becoming academically involved by participating in society simulation activities for higher learning.